Hewam S. Abera

20674665

1. Chapter 5, Section 5-1, Exercise 6

Circuit (a)

Boolean expression: X = (A + B)(C + D)

A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Circuit (b)

A	B	C	D	Y_1	Y_2	Y_3	X
0	0	0	0	1	1	1	0
0	0	0	1	1	1	1	0
0	0	1	0	1	0	1	0
0	0	1	1	1	0	0	1
0	1	0	0	1	1	1	0
0	1	0	1	1	1	1	0

Hewam S. Abera

20674665

0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	1	1	0
1	0	0	1	1	1	1	0
1	0	1	0	1	0	1	0
1	0	1	1	1	0	0	1
1	1	0	0	0	1	1	0
1	1	0	1	0	1	1	0
1	1	1	0	0	1	1	0
1	1	1	1	0	1	0	0

Circuit (c)

Boolean expression: X = ((AB) + C)D + E

A	B	C	D	E	X
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	0
0	0	0	1	1	1
0	0	1	0	0	0
0	0	1	0	1	1
0	0	1	1	0	1
0	0	1	1	1	1
0	1	0	0	0	0
0	1	0	0	1	1
0	1	0	1	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	0	1	1
0	1	1	1	0	1
0	1	1	1	1	1
1	0	0	0	0	0

Hewam S. Abera

20674665

1	0	0	0	1	1
1	0	0	1	0	0
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	0	1	1
1	0	1	1	0	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	0	1	1
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	0
1	1	1	0	1	1
1	1	1	1	0	1
1	1	1	1	1	1

Circuit (d)

A	B	C	D	Y_1	Y_2	Y_3	Y_4	X
0	0	0	0	1	1	0	1	0
0	0	0	1	1	1	0	0	1
0	0	1	0	1	1	0	1	0
0	0	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	1
0	1	0	1	0	1	1	0	1
0	1	1	0	0	0	1	0	1
0	1	1	1	0	0	1	0	1

Hewam S. Abera

20674665

1	0	0	0	1	1	0	1	0
1	0	0	1	1	1	0	0	1
1	0	1	0	1	1	0	1	0
1	0	1	1	1	1	0	0	1
1	1	0	0	1	1	0	1	0
1	1	0	1	1	1	0	0	1
1	1	1	0	1	0	1	0	1
1	1	1	1	1	0	1	0	1

Circuit (e)Circuit (e)

Boolean expression: $X = D(AB + C') + E'$

Inputs: A, B, C, D, E (32 combinations).

A	B	C	D	E	X
0	0	0	0	0	1
0	0	0	0	1	0
0	0	0	1	0	1
0	0	0	1	1	1
0	0	1	0	0	1
0	0	1	0	1	0
0	0	1	1	0	1
0	0	1	1	1	0
0	1	0	0	0	1
0	1	0	0	1	0
0	1	0	1	0	1
0	1	0	1	1	1
0	1	1	0	0	1
0	1	1	0	1	0
0	1	1	1	0	1
0	1	1	1	1	0
1	0	0	0	0	1

Hewam S. Abera

20674665

1	0	0	0	1	0
1	0	0	1	0	1
1	0	0	1	1	1
1	0	1	0	0	1
1	0	1	0	1	0
1	0	1	1	0	1
1	0	1	1	1	0
1	1	0	0	0	1
1	1	0	0	1	0
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	1
1	1	1	0	1	0
1	1	1	1	0	1
1	1	1	1	1	1

2. Chapter 5, Section 5-2, Exercise 13

The output is 1 for every input combination with C = 0, and also for A = B = C = 1. Thus X = C′ + AB. (Check: when C=1, output reduces to AB.)
Circuit: invert C, AND A with B, then OR those two results. This gives the listed outputs 1,0,1,0,1,0,1,1 in ABC order.

3. Chapter 5, Section 5-3, Exercise 20

(a)

Boolean expression: X = (NOT A)B + AC

- N1 = A NAND A = NOT A
- N2 = N1 NAND B
- N3 = A NAND C
- X = N2 NAND N3

(b)

Boolean expression: X = NOT[(NOT A)B + BC + C(NOT D)]

- N1 = A NAND A = NOT A
- N2 = D NAND D = NOT D
- N3 = N1 NAND B
- N4 = B NAND C
- N5 = C NAND N2
- N6 = N3 NAND N4
- X = N6 NAND N5

Hewam S. Abera

20674665

4. Chapter 6, Section 6-2, Exercise 4 — Parallel adder

Stage	A	B	Cin	Cout / Σ
Σ1 (LSB)	1	1	0	Cout=1, Σ1=0
Σ2	1	1	1	Cout=1, Σ2=1
Σ3	1	0	1	Cout=1, Σ3=0
Σ4 (MSB)	1	1	1	Cout=1, Σ4=1

The four sum bits are Σ4Σ3Σ2Σ1 = 1010, with final carry Σ4-output = 1. Complete result: 11010_2.

+5. 2-Bit Adder — Sum and Carry

Sum: S = A XOR B XOR Cin

Carry out: Cout = AB + ACin + BCin

Interval	A_1A_2	B_1B_2	Cin	Carry out	Sum (S_1S_2)
1	10	10	0	1	00
2	11	11	0	1	10
3	01	00	0	0	01
4	00	01	0	0	01
5	11	00	1	1	00
6	11	01	1	1	01
7	11	00	1	1	00
8	10	11	1	1	10

6. 4-Bit Comparator

Interval	A	B	A > B	A = B	A < B
1	9	4	1	0	0
2	15	15	0	1	0
3	14	2	1	0	0
4	12	3	1	0	0
5	5	12	0	0	1

7. Decoder Binary Codes

The most significant bit (MSB) is A_3.

- (a) 1110
- (b) 1100
- (c) 1111
- (d) 1000

Hewam S. Abera

20674665

8. Multiplexer Output

Given $S_1 = 0$ and $S_0 = 1$, input D_1 is selected.

$Y = D_1 = 1$